Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-D Valvular Heart Diseases

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-18-04-C01-BD1

Brain dump 1 · foundational retrieval: Rebuild calcific aortic stenosis and the wall it builds

A competent response contains

· The normal aortic valve has three cusps and an area of about 3 to 4 cm2; symptoms usually begin when the area falls to about 1 cm2 or less, so years pass silently.

· Calcific aortic stenosis is an active process: mechanical stress injures the cusp endothelium, low-density lipoprotein and lipoprotein(a) enter the cusp, inflammation follows, and valve interstitial cells become osteoblast-like and deposit calcium.

· Heaped calcified masses bulge into the sinuses of Valsalva and the commissures stay unfused.

· The calcification is dystrophic, so serum calcium is normal, and the risk factors match those of atherosclerosis.

· Age-related calcific disease on a once-normal valve usually presents after about age 70; a congenitally bicuspid valve, present in about 1 to 2% of people, concentrates stress and presents one to two decades earlier.

· Rheumatic aortic stenosis shows fused commissures and almost always accompanies rheumatic mitral disease.

· Law of Laplace: wall stress is proportional to pressure times radius divided by twice the wall thickness, so a rise in pressure is offset by a thicker wall.

· Pushing a normal stroke volume through a narrow orifice demands a much higher ventricular pressure, so myocytes add sarcomeres in parallel and the wall thickens around a normal or small cavity, which is concentric hypertrophy.

· Ejection fraction is preserved for years, but the stiff wall depends on atrial contraction for filling, which produces an S4, and demand eventually outgrows the capillary supply, giving fibrosis and failure.

Distinctions that matter

· Pressure load gives concentric hypertrophy with a normal cavity, while volume load gives eccentric hypertrophy with a dilated cavity.

· Unfused commissures with heaped nodular calcium point to calcific or bicuspid disease; fused commissures point to rheumatic disease.

Common misconceptions

· Calling calcific aortic stenosis simple mechanical wear. It is an inflammatory, lipid-driven, bone-forming process in the cusp.

· Expecting a raised serum calcium. Dystrophic calcification occurs in injured tissue with normal serum calcium.

Where it is taught: Section C01, first three paragraphs of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C01-BD2

Brain dump 2 · integration and application: Work a Doppler study back to the symptoms

A competent response contains

· Peak gradient is about 4 times velocity squared: 4 times 4.6 squared is about 85 mm Hg.

· With an arterial systolic pressure of 128 mm Hg the ventricle must generate about 213 mm Hg.

· By Laplace with an unchanged radius, thickness must rise in proportion to pressure: 213 divided by 128 is about 1.7, so a 1.0 cm wall must reach about 1.7 cm.

· This study is severe on all three thresholds: area about 1 cm2 or less (0.8 cm2), peak velocity about 4 m/s or more (4.6 m/s), and mean gradient about 40 mm Hg or more.

· Angina with clean coronary arteries: demand is high in a thick wall, a raised ventricular diastolic pressure compresses subendocardial vessels, and prolonged systole shortens diastolic perfusion time.

· Exertional syncope: exercise lowers peripheral resistance while output across a fixed orifice cannot rise, with a ventricular mechanoreceptor reflex and arrhythmias adding to it.

· Dyspnea: high ventricular diastolic pressure is transmitted back to the pulmonary veins.

· The carotid upstroke is weak, slow rising, and late peaking, which is pulsus parvus et tardus, because the orifice limits the rate of rise of aortic pressure; the apical impulse is sustained because a thick wall contracts long and hard.

· A failing ventricle generates less flow, so the murmur becomes softer even as the lesion worsens, which is why loudness does not track severity.

Distinctions that matter

· Jet velocity and gradient report the pressure difference, while valve area reports the orifice itself; a low-flow state can lower the gradient without the orifice improving.

· Angina here reflects supply and demand in a hypertrophied wall rather than fixed coronary obstruction.

Common misconceptions

· Reading a quiet murmur as mild disease. Reduced flow from a failing ventricle softens it.

· Expecting symptoms early. The valve area must fall to roughly a third of normal before the disease declares itself.

Where it is taught: Section C01, Doppler paragraph, findings table, and the worked example of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C01-Q01

Board-style question: Chest pain with clean coronary arteries

Correct answer: C. High oxygen demand in a thick wall whose raised diastolic pressure compresses subendocardial vessels while prolonged systole shortens diastolic perfusion time

Why it is right: Concentric hypertrophy raises oxygen demand, the raised left ventricular diastolic pressure compresses subendocardial vessels, and prolonged ejection shortens diastolic perfusion time, so the subendocardium becomes ischemic even though the epicardial arteries are normal.

A is wrong: Plaque rupture with thrombotic occlusion would produce an acute coronary syndrome, and angiography in this patient shows no epicardial stenosis.

B is wrong: Calcific embolism into a coronary artery is uncommon and would cause an abrupt infarct rather than months of predictable exertional tightness.

D is wrong: Loss of large von Willebrand factor multimers from high shear favours bleeding from intestinal angiodysplasia and does not reduce oxygen carrying capacity or cause angina.

A competent response contains

· The murmur, the weak late carotid upstroke, the soft A2, and the valve area of 0.8 cm2 with a mean gradient of 48 mm Hg establish severe aortic stenosis.

· Angina occurs with normal coronary arteries because demand is high in a 1.6 cm wall, raised diastolic pressure compresses subendocardial vessels, and prolonged systole shortens the time available for diastolic perfusion.

Distinctions that matter

· Ischemia in aortic stenosis is a supply and demand problem within the wall, not a problem of epicardial coronary obstruction.

Common misconceptions

· Assuming exertional chest pain always means coronary atherosclerosis. Severe aortic stenosis produces angina with clean arteries.

Where it is taught: Section C01 of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C02-BD1

Brain dump 1 · foundational retrieval: Causes of aortic regurgitation and the wide pulse pressure

A competent response contains

· Root causes: hypertension and aging, Marfan syndrome, tertiary syphilis, ankylosing spondylitis, and ascending aortic dissection; root disease also carries a risk of aneurysm and dissection.

· Leaflet causes: bicuspid valve, endocarditis perforating a cusp, and rheumatic cusp retraction, usually with mitral disease.

· Marfan syndrome: mutant fibrillin-1 (FBN1) releases excess TGF-beta activity, producing cystic medial degeneration.

· Tertiary syphilis: endarteritis of the vasa vasorum scars the aortic media and gives the intima a tree-bark appearance.

· In chronic aortic regurgitation the ventricle receives atrial inflow plus the regurgitant volume, so end-diastolic volume rises and the Frank-Starling mechanism raises total stroke volume; ejecting that volume into a high-pressure aorta adds a pressure load to the volume load.

· Sarcomeres are added in series, the chamber dilates with proportional wall growth (eccentric hypertrophy), and chronic aortic regurgitation produces some of the largest ventricles seen.

· The dilated chamber is compliant, so diastolic pressure stays near normal and patients can be asymptomatic for years until dilation outruns hypertrophy and contractility falls.

· A representative pressure is 170/45 mm Hg: systolic rises with the large rapidly ejected stroke volume, and diastolic falls because the aorta empties both backward into the ventricle and forward into vasodilated tissues.

· Named signs, all from the same wide pulse pressure: water-hammer or Corrigan pulses, head bobbing (de Musset), nail bed capillary pulsation (Quincke), pistol-shot femoral sound (Traube), and a to-and-fro femoral bruit on compression (Duroziez); the apical impulse is hyperdynamic and displaced down and laterally.

· The murmur is high-pitched, blowing, early diastolic, and decrescendo, beginning at A2 where the aorta to ventricle gradient is greatest, heard best at the left sternal border with the patient sitting forward in held expiration, and louder with handgrip because handgrip raises aortic diastolic pressure.

· Low aortic diastolic pressure lowers coronary perfusion pressure and can cause angina.

Distinctions that matter

· Root disease pulls normal cusps apart, while leaflet disease damages the cusps themselves; the distinction decides whether the aorta itself is also at risk.

· Handgrip raises afterload, so it makes aortic regurgitation louder while making the aortic stenosis gradient and murmur smaller.

Common misconceptions

· Treating the named peripheral signs as separate phenomena. Each one is the same wide pulse pressure seen in a different tissue.

· Placing the murmur in mid diastole. It begins at A2, when the pressure difference is largest, and decreases as the two pressures converge.

Where it is taught: Section C02, cause paragraphs and the sections on chronic loading, wide pulse pressure, and the murmur of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C02-BD2

Brain dump 2 · integration and application: Two patients with a leaking aortic valve

A competent response contains

· Patient 1 has a large, dilated, compliant ventricle: diastolic pressure stays near normal, the murmur is long because the gradient persists through diastole, and the pulse pressure is wide with bounding pulses.

· Patient 2 has a normal-sized, stiff ventricle: diastolic pressure rises steeply, aortic and ventricular pressures equalise quickly so the murmur is short and soft, and the pulse pressure is not wide.

· Compliance of the receiving chamber, not regurgitant volume alone, decides whether the patient compensates.

· A severe acute leak can be almost silent because the pressures equalise early in diastole, leaving little gradient and little turbulence to hear.

· The steeply rising ventricular diastolic pressure can close the mitral valve early, and pulmonary edema and shock follow.

· Regurgitation occurs only in diastole, so bradycardia lengthens diastole and increases the regurgitant volume per beat, while tachycardia reduces it; a slower rate therefore worsens the load in both patients and is particularly poorly tolerated in the acute lesion.

· An Austin Flint murmur is an apical mid-to-late diastolic rumble caused by the regurgitant jet partly closing the anterior mitral leaflet; mitral stenosis adds an opening snap and a loud S1, which the Austin Flint murmur lacks.

Distinctions that matter

· Murmur length in regurgitation reports how long a pressure gradient lasts, not how much blood leaks.

· A wide pulse pressure is a feature of the chronic lesion only, because it requires a large total stroke volume that a stiff ventricle cannot deliver.

Common misconceptions

· Expecting a wide pulse pressure in acute aortic regurgitation. Stroke volume is not large and the pressures equalise, so the pulse pressure is normal or narrow.

· Judging severity from how loud or long the murmur is. The quietest murmurs in this chapter belong to the most unstable patients.

Where it is taught: Section C02, murmur paragraph and acute aortic regurgitation paragraph, with the chronic against acute table in section C04 of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C03-BD1

Brain dump 1 · foundational retrieval: Mitral stenosis: the chambers that pay and the sounds you hear

A competent response contains

· Normal mitral valve area is about 4 to 6 cm2, so left atrial and ventricular pressures are nearly equal in diastole; a gradient appears below about 2.5 cm2 and severe disease is about 1.5 cm2 or less.

· Mitral stenosis is almost always rheumatic; heavy mitral annular calcification in older adults and congenital malformation are uncommon causes.

· The left atrium hypertrophies and dilates while the left ventricle is underfilled, normal or small, with normal or low diastolic pressure.

· The hallmark is a pulmonary capillary wedge pressure higher than left ventricular diastolic pressure.

· Hemoptysis comes from rupture of bronchial veins that connect with congested pulmonary veins.

· Loud S1: high left atrial pressure holds the leaflets wide open until systole begins, and S1 softens once they calcify.

· Opening snap after S2: fused leaflets dome open abruptly when ventricular pressure falls below atrial pressure.

· The interval from S2 to the opening snap shortens as the disease worsens, because higher left atrial pressure opens the valve sooner after A2.

· Low-pitched apical mid-diastolic rumble from turbulent flow through the orifice, heard with the bell in the left lateral decubitus position; its length rather than its loudness tracks severity.

· Presystolic accentuation comes from atrial contraction boosting flow and is lost in atrial fibrillation.

· Electrocardiogram: a broad notched P wave in lead II with a deep terminal negative P in V1 (P mitrale) from left atrial enlargement, and right axis deviation from pulmonary hypertension with right ventricular hypertrophy.

· The enlarged left atrium, the most posterior chamber, can indent the esophagus causing dysphagia, compress the left recurrent laryngeal nerve causing hoarseness (Ortner syndrome), and elevate the left main bronchus, widening the carinal angle.

Distinctions that matter

· The opening snap is diastolic and belongs to mitral stenosis; the mid-systolic click belongs to mitral valve prolapse.

· Murmur length tracks severity in mitral stenosis, while loudness does not.

Common misconceptions

· Expecting a dilated left ventricle. The obstruction sits upstream, so the ventricle is underfilled and small.

· Reading a shorter S2 to opening snap interval as milder disease. A shorter interval means a higher left atrial pressure and worse stenosis.

Where it is taught: Section C03, hemodynamics paragraph and the findings table of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C03-BD2

Brain dump 2 · integration and application: Why a faster heart rate floods the lungs

A competent response contains

· Flow across an orifice varies with its area times the square root of the pressure difference, so the gradient rises with the square of the flow rate.

· The ventricle fills only in diastole and tachycardia shortens diastole more than systole, so the same volume must cross the valve faster and left atrial pressure rises sharply; the notes give a rise from 70 to 120 per minute as roughly doubling the gradient.

· Her precipitants are fever, tachycardia, and pregnancy; exercise, anemia, and atrial fibrillation do the same in other patients.

· Pregnancy adds blood volume, cardiac output, and heart rate, each of which raises flow across the fixed orifice.

· Her left ventricular end-diastolic pressure and ventricular size are normal or low, because the obstruction lies upstream of the ventricle and the chamber is underfilled rather than overloaded.

· Atrial fibrillation removes the atrial contribution to filling and shortens diastole, so filling worsens twice over.

· Stasis in the left atrial appendage forms mural thrombi that embolise to the brain, kidneys, spleen, and limbs.

· In mitral stenosis the wedge pressure exceeds the left ventricular diastolic pressure, whereas in pulmonary edema from left ventricular failure the raised wedge pressure reflects an equally raised ventricular diastolic pressure.

Distinctions that matter

· Pulmonary edema upstream of a stenotic mitral valve coexists with a normal ventricular filling pressure; pulmonary edema from ventricular failure does not.

· Heart rate matters here through diastolic filling time, which is why rate control changes the gradient without changing the valve.

Common misconceptions

· Assuming any patient with pulmonary edema has a failing left ventricle. In mitral stenosis the ventricle is protected.

· Treating atrial fibrillation as a rhythm problem alone. It also removes atrial kick and shortens diastole, which is why it so often precipitates the first episode of edema.

Where it is taught: Section C03, paragraphs on heart rate and on atrial fibrillation, and the stop and predict box of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C03-RT01

Retrieval exercise: Rebuild the backward transmission of pressure

A competent response contains

· Step 1: fused, narrowed mitral orifice.

· Step 2: left atrial pressure rises to drive flow into the ventricle.

· Step 3: the left atrium hypertrophies and dilates while the left ventricle stays small.

· Step 4: pulmonary venous pressure rises, giving dyspnea, edema, and hemoptysis.

· Step 5: reactive arteriolar constriction and then remodelling of the arteriolar walls produce pulmonary hypertension.

· Step 6: right ventricular hypertrophy and dilation follow, and the dilated right ventricle stretches the tricuspid annulus so tricuspid regurgitation appears.

· Step 7: right heart failure, with distended neck veins, hepatomegaly, and peripheral edema.

· Pressures are raised in the pulmonary veins, pulmonary capillaries, pulmonary arteries, right ventricle, right atrium, and left atrium, and are normal or low in the left ventricle.

· The left ventricle is spared because it sits downstream of the obstruction and simply receives less blood.

Distinctions that matter

· Pulmonary hypertension here begins as reversible arteriolar constriction and becomes fixed once the walls remodel.

· Tricuspid regurgitation in this sequence is functional, caused by right ventricular dilation rather than by disease of the tricuspid leaflets.

Common misconceptions

· Drawing the left ventricle as loaded. Every chamber upstream of the narrowed valve carries the load and the ventricle does not.

Where it is taught: Section C03, backward transmission flow diagram of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C04-BD1

Brain dump 1 · foundational retrieval: The mitral apparatus and how each part fails

A competent response contains

· The apparatus is the annulus, leaflets, chordae, papillary muscles, and the ventricular wall that holds them.

· Primary mitral regurgitation: mitral valve prolapse, which is the most common primary cause in developed countries, rheumatic disease, endocarditis, and papillary muscle or chordal rupture.

· Secondary or functional mitral regurgitation: a dilated ventricle stretches the annulus and displaces the papillary muscles, tethering normal leaflets; ischemic papillary muscle dysfunction causes transient regurgitation.

· The posteromedial papillary muscle ruptures most often after infarction because it usually has a single arterial supply.

· Functional regurgitation feeds itself: the added regurgitant volume dilates the ventricle further, which stretches the annulus and tethers the leaflets more.

· Left atrial pressure is low, so backflow begins at the onset of systole before the aortic valve opens, and there is no true isovolumetric contraction.

· The regurgitant volume returns to the ventricle in early diastole, so both the atrium and the ventricle are volume loaded.

· The left atrium dilates and becomes compliant, holding a large volume at modest pressure, and often fibrillates; the ventricle develops eccentric hypertrophy.

· The murmur is holosystolic, high-pitched, and blowing, running from S1 through A2, loudest at the apex and radiating to the left axilla.

· Handgrip and squatting make it louder, Valsalva and standing soften it, and a ventricular septal defect murmur is heard at the left lower sternal border instead.

· S1 is soft, early A2 widens the split of S2, and rapid early diastolic inflow from the loaded atrium produces an S3 that reflects volume rather than proving failure.

Distinctions that matter

· Primary disease is of the leaflets or chordae; secondary disease is of the ventricle, with normal leaflets tethered out of position.

· An S3 in chronic mitral regurgitation reports a large early diastolic inflow, so it is not by itself evidence of heart failure.

Common misconceptions

· Expecting an isovolumetric phase in mitral regurgitation. Blood escapes into the atrium as soon as ventricular pressure exceeds atrial pressure.

· Attributing all mitral regurgitation to leaflet disease. A dilated ventricle alone can make normal leaflets incompetent.

Where it is taught: Section C04, cause paragraph, loading paragraph, and murmur paragraph of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C04-BD2

Brain dump 2 · integration and application: Why a normal ejection fraction misleads

A competent response contains

· Total stroke volume is 210 minus 75, which is 135 mL.

· Ejection fraction is 135 divided by 210, which is about 64%, a value that looks normal.

· Regurgitant volume is 60% of 135, which is 81 mL, so forward stroke volume is 135 minus 81, which is 54 mL.

· With end-systolic volume at 105 mL, total stroke volume is 105 mL, ejection fraction is 105 divided by 210, which is 50%, regurgitant volume is 63 mL, and forward stroke volume is 42 mL.

· The ventricle unloads into a low-pressure atrium, so ejection fraction overstates contractility and a low-normal value already signals impairment; the chapter notes about 60% or less as the guideline threshold suggesting dysfunction in chronic mitral regurgitation.

· In aortic stenosis the ventricle ejects against a high pressure, so a preserved ejection fraction is a more honest report of contractility and a fall in it indicates late disease.

· Forward stroke volume is the measurement that reports how much blood actually reaches the tissues.

Distinctions that matter

· Total stroke volume and forward stroke volume are different quantities, and only the second one perfuses the body.

· Ejection fraction depends on the load the ventricle ejects against, which is why the same number means different things in mitral regurgitation and in aortic stenosis.

Common misconceptions

· Reading an ejection fraction of 50% in chronic mitral regurgitation as mildly reduced. In this lesion it reports substantial loss of contractility.

· Assuming a large total stroke volume means good cardiac output. Most of it may be moving backward.

Where it is taught: Section C04, worked example on ejection fraction, with the threshold noted in the verify list of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C04-CP01

Comparison table: Chronic against acute regurgitation

A competent response contains

· Receiving chamber: chronic disease dilates it and makes it compliant, the ventricle in aortic regurgitation and both the atrium and the ventricle in mitral regurgitation; in acute disease it is normal in size and stiff.

· Filling pressure: chronic disease keeps it near normal or moderately raised for years; acute disease raises it steeply, as left ventricular diastolic pressure in aortic regurgitation and as left atrial pressure with tall v waves in mitral regurgitation.

· Murmur: chronic disease gives a long murmur, early diastolic and decrescendo in aortic regurgitation and holosystolic in mitral regurgitation; acute disease gives a short, soft murmur that is sometimes nearly absent.

· Pulse pressure in aortic regurgitation: wide with bounding pulses when chronic, not wide when acute.

· Presentation: chronic disease is silent for years and then brings exertional dyspnea and failure; acute disease brings sudden pulmonary edema and shock.

· Typical causes: chronic aortic regurgitation from root dilation, a bicuspid valve, or rheumatic disease, and chronic mitral regurgitation from prolapse, rheumatic disease, or ventricular dilation; acute disease from endocarditis, aortic dissection, or trauma, and in the mitral valve from papillary muscle or chordal rupture.

· Compliance of the receiving chamber, not regurgitant volume alone, decides whether the patient compensates.

Distinctions that matter

· Acute and chronic lesions can leak the same volume and look completely different, because a compliant chamber absorbs the volume at low pressure.

· A short soft murmur in regurgitation suggests early equalisation of pressures, which marks the acute and more dangerous lesion.

Common misconceptions

· Grading severity by murmur intensity in regurgitation. Loudness tracks the gradient and the duration of flow, not the volume that leaks.

· Expecting bounding pulses in every patient with aortic regurgitation. They require the large stroke volume of the chronic lesion.

Where it is taught: Section C04, chronic against acute regurgitation table, with the acute paragraphs in C02 and C04 of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C04-CA01

Clinical application: Sudden breathlessness four days after an infarction

A competent response contains

· Recognize: acute severe mitral regurgitation from rupture of the posteromedial papillary muscle, part of the mitral apparatus, leaving a flail posterior leaflet.

· Explain: the left atrium is normal in size and stiff, so the regurgitant volume raises atrial pressure steeply, producing tall v waves and a wedge pressure of 34 mm Hg that is transmitted straight to the pulmonary capillaries, giving edema within minutes; a chronically dilated compliant atrium would have absorbed the same volume at modest pressure.

· Differentiate: chronic mitral regurgitation has a dilated atrium and ventricle, a long loud holosystolic murmur radiating to the axilla, an S3, and years of exertional symptoms; acute aortic regurgitation gives a short early diastolic murmur with a steeply raised ventricular diastolic pressure and no tall atrial v waves, and the cardiac silhouette here is normal because there has been no time to dilate.

· Investigate: on echocardiography look for a flail posterior leaflet with a ruptured chordal or papillary attachment, a normal-sized left atrium and left ventricle, inferior wall hypokinesis, and a severe regurgitant jet; on the catheter tracing look for tall systolic v waves on the wedge trace with a high mean wedge pressure.

· Decide: the murmur is soft and short because left atrial pressure rises so high during systole that it approaches ventricular pressure late in systole, leaving little gradient and little turbulence, so nothing about severity can be inferred from its loudness.

· Justify: the posteromedial papillary muscle usually has a single arterial supply, which makes it vulnerable in an inferior infarct supplied by the artery that was occluded, and the late treatment allowed full necrosis; a small stiff atrium cannot accommodate the regurgitant volume, so pressures equalise late in systole and pulmonary pressures rise instead.

Distinctions that matter

· The normal cardiac silhouette and normal chamber sizes with severe regurgitation date the lesion as acute.

· Tall v waves on the wedge tracing localise the problem to a stiff left atrium receiving systolic backflow.

Common misconceptions

· Taking a soft murmur as evidence of a mild leak in a shocked patient. In acute mitral regurgitation the most severe lesions can be nearly silent.

· Expecting the anterolateral papillary muscle to rupture. The posteromedial muscle usually has one arterial supply and is the one that fails.

Where it is taught: Section C04, cause paragraph, acute mitral regurgitation paragraph, and the chronic against acute table of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C05-BD1

Brain dump 1 · foundational retrieval: How a vegetation forms and what the organism decides

A competent response contains

· A vegetation is a mass of platelets and fibrin with organisms and inflammatory cells attached to a valve or the endocardium; organisms grow inside an avascular fibrin mesh that neutrophils reach poorly, so the vegetation sheds them continuously.

· Pathogenesis in order: turbulence or prior damage injures valve endothelium; platelets and fibrin form a sterile nidus; transient bacteremia delivers organisms; adhesins attach them to the deposit; organisms multiply under fibrin, shielded from phagocytes; the vegetation grows, destroys tissue, and sheds emboli and organisms.

· Virulent Staphylococcus aureus can invade intact endothelium and skip the first two steps, which is why it infects normal valves.

· Collagen and tissue factor are exposed by turbulence, a scarred or congenitally abnormal valve, a prosthesis, or a catheter; transient bacteremia follows chewing, dental work, procedures, injection drug use, or skin infection.

· Viridans streptococci attach through adhesins and dextran; S. aureus binds fibrinogen and fibronectin through surface proteins such as clumping factor.

· Acute disease: S. aureus, usually a normal valve, days, high fever with rigors, large destructive vegetations with ring abscesses and metastatic infection, and embolic signs prominent; portals are skin, injection drug use, and catheters.

· Subacute disease: viridans streptococci of low virulence, a previously damaged valve (rheumatic, congenital, bicuspid, or prolapse with regurgitation), weeks to months, low-grade fever with fatigue, weight loss, anemia, splenomegaly, and clubbing when long-standing, with immunologic signs dominating because prolonged bacteremia allows immune complexes to form.

· Enterococci after genitourinary or gastrointestinal procedures and Streptococcus gallolyticus with colon cancer usually follow a subacute course.

· Left-sided valves face higher pressures and turbulent jets, so the mitral valve is most often involved, closely followed by the aortic.

· In people who inject drugs the tricuspid valve is most often involved because skin organisms and injected particles reach it first, with S. aureus dominating and Pseudomonas and Candida less often.

· Right-sided vegetations embolise to the lungs as multiple, often cavitating septic emboli; left-sided fragments cause infarcts and abscesses in the brain, spleen, kidneys, and limbs and seed arterial walls as mycotic aneurysms.

Distinctions that matter

· Virulence sets the tempo: a destructive organism needs no abnormal valve, while a low-virulence organism needs one.

· Side of the heart predicts where the emboli land, which is why pulmonary lesions point to a right-sided vegetation.

Common misconceptions

· Assuming endocarditis always requires a damaged valve. S. aureus invades intact endothelium.

· Expecting immunologic signs in acute disease. Immune complexes need weeks of bacteremia to form, so they belong to subacute disease.

Where it is taught: Section C05, pathogenesis flow diagram and the paragraphs on virulence and on valve involvement of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C05-BD2

Brain dump 2 · integration and application: Read the peripheral signs and the laboratory panel

A competent response contains

· Painful red-purple nodules on the finger pads are Osler nodes, caused by immune complex small vessel vasculitis and more typical of subacute disease.

· Painless flat red or hemorrhagic macules on the palms are Janeway lesions, caused by septic microemboli with small dermal abscesses and more typical of acute disease.

· Retinal hemorrhages with a pale centre are Roth spots, caused by a fibrin-platelet plug from immune vasculitis or microemboli, and painless.

· The sorting rule is that embolic lesions are painless while immune complex lesions hurt, because the vessel wall itself is inflamed.

· Red cell casts with low complement are immune complex glomerulonephritis from deposition in the glomeruli, and her 7-week illness is exactly the prolonged bacteremia that allows immune complexes to form.

· Her anemia is anemia of chronic inflammation, driven by hepcidin, alongside a raised erythrocyte sedimentation rate and C-reactive protein; rheumatoid factor may be positive in subacute disease.

· Continuous shedding of organisms makes the bacteremia persistent, so cultures drawn at different times grow the same organism, which tells you the vegetation itself is seeding the blood.

· A subacute course on a previously abnormal aortic valve points to viridans streptococci.

· Duke criteria: positive blood cultures and echocardiographic evidence of a vegetation, abscess, or new regurgitation are major criteria; a predisposing valve condition, fever of 38.0 °C (100.4 °F) or higher, and embolic or immunologic phenomena are minor criteria, so her cultures are major and her valve, fever, Osler nodes, Janeway lesions, Roth spots, glomerulonephritis, and rheumatoid factor fall among the minor criteria.

Distinctions that matter

· Osler nodes and Janeway lesions sit on the same hands and separate by pain, which follows from vasculitis against embolism.

· Persistent bacteremia with the same organism in separated cultures is what distinguishes endocarditis from a transient bacteremia.

Common misconceptions

· Treating the peripheral signs as specific. Splinter hemorrhages and conjunctival petechiae in particular are nonspecific.

· Explaining the anemia as blood loss or hemolysis. It is the hepcidin-driven anemia of chronic inflammation.

Where it is taught: Section C05, peripheral signs table and the final paragraph on cultures and criteria of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C05-Q02

Board-style question: A painful nodule on the finger pad

Correct answer: A. Immune complex deposition with small vessel vasculitis in the digital pad

Why it is right: Weeks of low-grade bacteremia with a low-virulence organism allow immune complexes to form and deposit in small vessels, and vasculitis of the digital vessels inflames the vessel wall, which is why these nodules are raised and tender.

B is wrong: Septic microemboli with small dermal abscesses produce Janeway lesions, which are flat and painless and sit on the palms and soles.

C is wrong: Fibrin and platelet plugs in retinal arterioles produce Roth spots, which are retinal hemorrhages with pale centres and cause no pain in the fingers.

D is wrong: A mycotic aneurysm is a weakened arterial wall seeded by organisms, typically in a larger artery, and it does not present as multiple tender nodules on finger pads.

A competent response contains

· A 6-week illness on a rheumatic valve with viridans streptococci is subacute endocarditis, in which prolonged bacteremia allows immune complexes to form.

· Tender nodules on the finger pads are Osler nodes from immune complex small vessel vasculitis, and the same mechanism explains her glomerulonephritis, low complement, and positive rheumatoid factor.

Distinctions that matter

· Painful lesions in endocarditis are immune complex lesions; painless lesions are embolic.

Common misconceptions

· Using Osler nodes and Janeway lesions interchangeably. One is painful and immune, the other painless and embolic.

Where it is taught: Section C05 of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C06-BD1

Brain dump 1 · foundational retrieval: The two sterile vegetations

A competent response contains

· Nonbacterial thrombotic endocarditis, also called marantic endocarditis, consists of small fibrin-platelet thrombi about 1 to 5 mm across, bland and loosely attached, along the lines of closure of otherwise normal mitral and aortic leaflets, with no organisms and little inflammation.

· It arises when hypercoagulability meets the normal wear at the line of closure.

· The key association is advanced malignancy, especially mucin-producing adenocarcinoma of the pancreas or lung; tumour tissue factor and mucins, which engage selectins on platelets and leukocytes, also cause migratory thrombophlebitis (Trousseau syndrome).

· Other settings are disseminated intravascular coagulation, sepsis, antiphospholipid syndrome, and indwelling catheters.

· There is usually no murmur, but fragments cause infarcts in the brain, kidneys, and spleen, and multiple strokes in different arterial territories in a patient with cancer are the classic presentation.

· A sterile lesion can also become the nidus for infective endocarditis.

· Libman-Sacks endocarditis occurs in systemic lupus erythematosus, often with antiphospholipid antibodies.

· Its warty vegetations are about 1 to 4 mm and can sit on either or both leaflet surfaces, including the ventricular undersurface of the mitral valve, and on the chordae and mural endocardium.

· They show fibrinoid necrosis with inflammation from immune complex injury, and healing fibrosis deforms the leaflets, most often causing mitral regurgitation.

· Emboli and the thrombotic tendency of antiphospholipid syndrome add further risk.

· Vegetations on both leaflet surfaces, rather than on the closure line alone, point to Libman-Sacks.

Distinctions that matter

· Both lesions are sterile, but only Libman-Sacks is inflammatory and only Libman-Sacks scars the leaflet enough to leave regurgitation.

· Position separates them: closure lines for nonbacterial thrombotic endocarditis, either or both surfaces for Libman-Sacks.

Common misconceptions

· Expecting a murmur. Nonbacterial thrombotic endocarditis usually produces none, because the leaflet is not deformed and nothing obstructs or leaks.

· Assuming a sterile vegetation stays sterile. It makes an excellent nidus for organisms.

Where it is taught: Section C06, paragraphs on nonbacterial thrombotic and Libman-Sacks endocarditis and the vegetation comparison table of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C06-BD2

Brain dump 2 · integration and application: Strokes in three territories with negative cultures

A competent response contains

· This is nonbacterial thrombotic endocarditis, pointed to by a mucin-producing pancreatic adenocarcinoma and by small bland nodules on the closure lines of normal leaflets with no fever and no murmur.

· Mechanism: tumour tissue factor and mucins engaging selectins on platelets and leukocytes create a hypercoagulable state, and fibrin-platelet thrombi form where the leaflets meet and wear.

· Infarcts appear in unrelated territories because fragments are loosely attached and shower repeatedly into the systemic circulation from a left-sided valve.

· There is no murmur because the vegetations are small and do not obstruct the orifice or prevent leaflet closure.

· Negative cultures do not prove sterility: fastidious organisms and prior antibiotics cause culture-negative infective endocarditis.

· Findings favouring infection: persistent fever, valve destruction, new regurgitation, and immunologic signs such as glomerulonephritis or Osler nodes.

· Findings keeping you with a sterile lesion: known cancer, lupus, or a coagulopathy, with bland infarcts and no destruction.

· If organisms seeded the nodules they would enlarge and become bulky and friable, destroy the leaflet with perforation or abscess, produce new regurgitation with a murmur, and shed septic rather than bland emboli.

Distinctions that matter

· Embolic burden can be identical in sterile and infected vegetations; destruction of the valve is what separates them.

· Multiple strokes in different arterial territories point to a cardiac or proximal aortic source showering repeatedly rather than to disease of one artery.

Common misconceptions

· Taking negative blood cultures as proof that a vegetation is sterile.

· Expecting a hypercoagulable patient with valve nodules to be febrile. Absence of fever is typical of the sterile lesion.

Where it is taught: Section C06, nonbacterial thrombotic endocarditis paragraph and the closing paragraph on culture-negative disease of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C06-CP02

Comparison table: Four kinds of valve vegetation

A competent response contains

· Size and consistency: infective vegetations are large, bulky, and friable; nonbacterial thrombotic about 1 to 5 mm; Libman-Sacks about 1 to 4 mm and warty; acute rheumatic fever about 1 to 2 mm warty nodules.

· Position: infective lesions sit on closure lines and can spread to chordae and the wall; nonbacterial thrombotic on the lines of closure; Libman-Sacks on either or both surfaces, chordae, and mural endocardium; rheumatic on the lines of closure.

· Organisms: present in infective disease, absent in the other three.

· Inflammation and damage: marked in infective disease with perforation and abscess; minimal with no damage in nonbacterial thrombotic disease; fibrinoid necrosis with scarring and mitral regurgitation in Libman-Sacks; part of a pancarditis in acute rheumatic fever, with scarring after recurrences.

· Emboli: septic in infective disease, bland and often causing multiple infarcts in nonbacterial thrombotic disease, bland in Libman-Sacks, and uncommon in acute rheumatic fever.

· Setting: bacteremia; cancer, disseminated intravascular coagulation, and other hypercoagulable states; lupus with antiphospholipid antibodies; and 2 to 4 weeks after group A streptococcal pharyngitis.

· Size and destruction separate infected from sterile vegetations, involvement of both leaflet surfaces points to Libman-Sacks, and the clinical setting separates nonbacterial thrombotic from rheumatic lesions.

Distinctions that matter

· Three of the four are sterile, so the discriminating features are size, destruction, position, and the clinical setting rather than the presence of a vegetation.

· Rheumatic and nonbacterial thrombotic vegetations both sit on closure lines and are small, which is why the history decides between them.

Common misconceptions

· Treating any vegetation as evidence of infection.

· Assuming a sterile vegetation leaves the valve normal. Libman-Sacks and rheumatic lesions both scar.

Where it is taught: Section C06, comparison of valve vegetations table, with the rheumatic lesion in C08 of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C07-BD1

Brain dump 1 · foundational retrieval: Myxomatous degeneration and the mid-systolic click

A competent response contains

· Mitral valve prolapse is systolic billowing of one or both mitral leaflets above the annulus into the left atrium, affecting about 2 to 3% of adults and usually asymptomatic.

· The lesion is myxomatous degeneration: the spongiosa expands with proteoglycans and glycosaminoglycans while the collagen of the fibrosa becomes thin and disorganised, driven by valve interstitial cells overproducing matrix-degrading enzymes.

· The leaflets become thick, redundant, and hooded between chordal attachments, the chordae elongate, thin, and may rupture, the posterior leaflet is most often involved, and the annulus may dilate.

· Marfan syndrome: deficient fibrillin-1 microfibrils fail to hold TGF-beta inactive, and excess signalling disrupts the valve matrix. Ehlers-Danlos syndromes: defective collagen weakens the fibrosa. Autosomal dominant polycystic kidney disease: abnormal polycystins, with berry aneurysms as an associated finding.

· Click mechanism: as the ventricle empties, the papillary muscles and annulus move closer together, the redundant leaflets lose support and billow into the atrium, and the slack chordae suddenly tighten, producing a sharp mid-systolic click at the apex.

· If coaptation then fails, a late systolic murmur of mitral regurgitation follows the click to S2.

· Complications follow from weak tissue: progressive mitral regurgitation as leaflets stretch, acute severe regurgitation when a thinned chorda ruptures and leaves a flail segment, infective endocarditis on prolapsing leaflets especially with regurgitation, platelet-fibrin thrombi on the atrial surface that embolise and cause transient ischemic attacks or stroke, and atrial and ventricular arrhythmias linked to papillary muscle traction and adjacent fibrosis; sudden cardiac death is rare.

Distinctions that matter

· Myxomatous degeneration adds proteoglycan and weakens collagen, which is the opposite of the fibrosis and fusion of rheumatic disease.

· The click comes from chordae snapping taut, while the murmur comes from leaflets failing to meet; a patient can have the click without the murmur.

Common misconceptions

· Expecting a thin, shrunken valve. The leaflets are thick, redundant, and hooded.

· Placing the click in early systole. It occurs in mid systole, once the ventricle has emptied enough for the leaflets to lose support.

Where it is taught: Section C07, first two paragraphs, the click paragraph, and the complications paragraph of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C07-BD2

Brain dump 2 · integration and application: Tell four systolic findings apart at the bedside

A competent response contains

· A smaller ventricle reaches the critical size sooner, so the click moves toward S1 and the murmur lengthens; a larger ventricle delays the click toward S2 and shortens the murmur.

· Smaller ventricle: standing and the strain phase of Valsalva. Larger ventricle: squatting, passive leg raise, and sustained handgrip.

· Mitral valve prolapse: standing and Valsalva move the click earlier and lengthen the murmur; squatting and handgrip delay the click and shorten the murmur, because the effect depends on chamber size.

· Hypertrophic cardiomyopathy: standing and Valsalva make the murmur louder and squatting and handgrip make it softer, because a smaller cavity worsens outflow obstruction.

· Aortic stenosis: standing and Valsalva make the murmur softer because less preload means less flow, and handgrip also softens it by raising afterload and lowering the gradient.

· Chronic mitral regurgitation: handgrip and squatting make it louder by raising afterload, and standing and Valsalva soften it by reducing preload.

· Prolapse and hypertrophic cardiomyopathy both worsen with a smaller ventricle; the mid-systolic click with a late systolic murmur at the apex separates prolapse from the crescendo-decrescendo murmur of outflow obstruction.

· The click is systolic and moves with ventricular size, while the opening snap of mitral stenosis is diastolic and moves with left atrial pressure.

· The patient with prolapse is the one at risk of sudden severe regurgitation, which requires a thinned chorda to rupture and leave a flail leaflet segment.

Distinctions that matter

· Manoeuvres separate lesions by whether the finding depends on flow across the valve or on the size of the cavity.

· Handgrip raises afterload, which increases regurgitant murmurs and the aortic regurgitation murmur while reducing the aortic stenosis gradient.

Common misconceptions

· Expecting everything to get louder with standing. Standing reduces preload, so flow murmurs soften while cavity-dependent findings intensify.

· Confusing the click with the opening snap. One is systolic and depends on ventricular size, the other diastolic and depends on atrial pressure.

Where it is taught: Section C07, manoeuvre paragraph, with the aortic stenosis findings table in C01 and the mitral regurgitation murmur paragraph in C04 of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C07-PR01

Prediction exercise: Predict what happens when a patient stands up

A competent response contains

· Standing reduces venous return, so left ventricular end-diastolic volume falls.

· The ventricle is smaller throughout systole, so it reaches the critical size at which the redundant leaflets lose support earlier in systole.

· The click moves earlier, closer to S1, because the chordae snap taut sooner.

· The late systolic murmur lengthens, because coaptation fails for a greater part of systole.

· The murmur of hypertrophic cardiomyopathy becomes louder, because a smaller cavity worsens dynamic outflow obstruction.

· The murmur of aortic stenosis becomes softer, because reduced preload means less flow across a fixed orifice.

Distinctions that matter

· Prolapse and hypertrophic obstruction both intensify with a smaller ventricle, while a fixed obstruction such as aortic stenosis depends on flow and therefore softens.

· The click changes its timing while the murmur changes its length, and both follow from the same change in chamber size.

Common misconceptions

· Expecting the click and the murmur to move in the same direction on the timeline. An earlier click leaves more of systole for the murmur, so the murmur gets longer.

· Assuming any murmur that softens on standing is aortic stenosis. Mitral regurgitation softens too, and the timing separates them.

Where it is taught: Section C07, manoeuvre paragraph and the stop and predict box of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C08-BD1

Brain dump 1 · foundational retrieval: The chronic rheumatic valve and which valves it takes

A competent response contains

· Acute rheumatic fever follows group A streptococcal pharyngitis by about 2 to 4 weeks.

· Antibodies and T cells against streptococcal antigens, particularly M protein, cross-react with cardiac myosin and valve tissue, which is molecular mimicry, causing a pancarditis with Aschoff bodies and tiny sterile vegetations along the lines of closure.

· A single attack may heal with little damage, but each new streptococcal throat infection can reinflame already scarred valves.

· Healing replaces inflamed tissue with fibrous tissue and brings new vessels into normally avascular leaflets, so damage accumulates and becomes evident years to decades after the first attack.

· The four changes of the chronic lesion: thickened, fibrotic, later calcified leaflets; fused commissures; thickened, shortened, fused chordae; and leaflet neovascularisation.

· Seen from the atrium the stenotic mitral valve has a rigid fish-mouth or buttonhole orifice, and Aschoff bodies are rarely found because scar has replaced them.

· Commissural fusion causes the stenosis, while retracted leaflets and short fused chordae prevent closure, so combined stenosis and regurgitation is common; degenerative disease by contrast usually causes isolated aortic stenosis or isolated mitral regurgitation.

· The mitral valve is involved in nearly all cases, alone in about two-thirds and with the aortic valve in about one-quarter; tricuspid involvement almost always accompanies left-sided disease and the pulmonic valve is rarely affected.

· The usual explanation for the left-sided preference is that left-sided valves close against systemic pressure and scar more after each episode.

· Rheumatic aortic disease almost never occurs without mitral disease, so isolated aortic disease in an older adult suggests a calcific or bicuspid valve.

· Rheumatic disease remains the leading cause of mitral stenosis worldwide, especially where streptococcal pharyngitis goes untreated.

Distinctions that matter

· Commissural fusion is the signature of rheumatic disease, while heaped nodular calcium with unfused commissures is the signature of degenerative disease.

· Acute rheumatic carditis is active inflammation; chronic rheumatic heart disease is the scar left behind.

Common misconceptions

· Describing the chronic valve as infected. The organism is long gone and the damage is immune scarring.

· Expecting Aschoff bodies in the chronic lesion. They are rarely found once fibrosis has replaced them.

Where it is taught: Section C08, all four paragraphs of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C08-BD2

Brain dump 2 · integration and application: Two narrowed valves with different histories

A competent response contains

· Patient 1 has chronic rheumatic mitral stenosis with aortic involvement, confirmed on the valve by fused commissures and thickened, shortened, fused chordae, with leaflet fibrosis and neovascularisation.

· Patient 2 has age-related calcific aortic stenosis, confirmed by heaped calcified masses bulging into the sinuses of Valsalva with unfused commissures.

· Rheumatic aortic disease almost never occurs without mitral disease, so a normal mitral valve effectively excludes it in patient 2.

· Commissures are fused in patient 1 and unfused in patient 2.

· Patient 1 can have both narrowing and leakage of the same valve because commissural fusion narrows the orifice while retracted leaflets and short fused chordae prevent the leaflets from meeting.

· She is at particular risk of subacute infective endocarditis caused by viridans streptococci, because her scarred valve exposes collagen and tissue factor and supports a sterile platelet-fibrin deposit that a low-virulence organism can colonise.

· Pregnancy brings more blood volume, higher cardiac output, and a faster heart rate; each raises flow across a fixed orifice, and because the gradient rises with the square of the flow rate and tachycardia shortens diastole, left atrial pressure climbs steeply and pulmonary edema follows.

· Classic decompensation in pregnancy is a recognised consequence of rheumatic mitral stenosis.

Distinctions that matter

· The pattern of valves involved is as informative as the appearance of any single valve, because rheumatic disease takes the mitral valve first.

· Rheumatic disease commonly gives stenosis and regurgitation together, while degenerative disease usually gives one lesion alone.

Common misconceptions

· Reading isolated aortic stenosis in an older adult as old rheumatic disease. Without mitral involvement it is calcific or bicuspid.

· Expecting a tight mitral valve to tolerate pregnancy because the patient has been well. Volume, output, and rate all rise and the orifice cannot.

Where it is taught: Section C08, final two paragraphs, with calcific disease in C01 and the heart rate paragraph in C03 of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C08-RT02

Retrieval exercise: Rebuild the path from a sore throat to a fused valve

A competent response contains

· Step 1: group A streptococcal pharyngitis.

· Step 2: molecular mimicry causes acute rheumatic carditis, with antibodies and T cells against M protein cross-reacting with cardiac myosin and valve tissue.

· Step 3: valvulitis with small vegetations about 1 to 2 mm on the lines of closure.

· Step 4: healing by fibrosis and neovascularisation; new vessels in a normally avascular leaflet give inflammatory cells continuing access, so each later infection can reach and reinflame the valve.

· Step 5: recurrent infections reinflame scarred valves, and each recurrence adds further fibrosis and deformity.

· Step 6: commissural fusion with thick, short, fused chordae.

· Step 7: fish-mouth mitral stenosis, often with mitral regurgitation and aortic disease.

· Chronic rheumatic heart disease is the scar of repeated immune injury, not an ongoing infection of the valve.

Distinctions that matter

· Neovascularisation is a healing change, not an acute one, and it is what makes an avascular leaflet accessible to later immune injury.

· Fusion of the commissures causes the stenosis, while shortening of the chordae prevents closure.

Common misconceptions

· Expecting damage after one attack. Accumulation across recurrences is what produces the chronic lesion.

· Calling the chronic valve infected and looking for organisms on it.

Where it is taught: Section C08, flow diagram from pharyngitis to chronic valve disease of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C09-BD1

Brain dump 1 · foundational retrieval: Serotonin, two filters, and the right-sided plaques

A competent response contains

· Carcinoid tumours are well-differentiated neuroendocrine tumours of enterochromaffin cells, most often in the ileum.

· Serotonin is made from tryptophan by tryptophan hydroxylase and aromatic amino acid decarboxylase; monoamine oxidase converts serotonin to 5-hydroxyindoleacetic acid, which is measured in the urine.

· Other mediators are histamine, kallikrein, which generates bradykinin, tachykinins, and prostaglandins.

· Sequence: a serotonin-secreting tumour, often ileal; metastasis to the liver; serotonin enters the hepatic veins, bypassing hepatic monoamine oxidase; high serotonin reaches the right heart; 5-HT2B signalling activates valve interstitial cells; fibrous plaques form on the tricuspid and pulmonic valves; lung monoamine oxidase lowers serotonin before the left heart.

· A tumour confined to the bowel drains through the portal vein, so hepatic monoamine oxidase inactivates its products and no valve disease follows; liver metastases release products straight into the hepatic veins and the right atrium.

· The lung protects the left heart because pulmonary endothelium takes serotonin up through its transporter and degrades it with monoamine oxidase.

· Left-sided lesions appear with a patent foramen ovale or other right-to-left shunt, with very high levels, or with a bronchial carcinoid draining into the pulmonary veins; ovarian carcinoids drain outside the portal system and can cause the syndrome without liver metastases.

· The lesions are pearly white plaques of myofibroblasts in glycosaminoglycan and collagen laid on right-sided endocardium, without destroying the valve beneath.

· Serotonin acts on 5-HT2B receptors of valve interstitial cells, partly through TGF-beta, to drive proliferation and matrix production; 5-HT2B agonists such as fenfluramine and certain ergot-derived dopamine agonists produce similar plaques.

· Tricuspid leaflets are fixed open, so tricuspid regurgitation is the most common lesion and tricuspid stenosis can also occur; pulmonic cusps thicken and retract, so pulmonic stenosis often coexists with regurgitation, and the right ventricle fails under combined volume and pressure loads.

Distinctions that matter

· The plaque sits on the surface and fixes the leaflet, which is why the valve beneath is intact and the lesion is not destructive.

· Two filters stand between the tumour and the left heart: the liver and the lung, and the lesion appears only when one of them is bypassed.

Common misconceptions

· Expecting valve disease from any carcinoid tumour. A bowel tumour without liver metastases usually causes none.

· Assuming a right-sided lesion means the organism or the lesion started there. The side affected is decided by which filter the products escape.

Where it is taught: Section C09, first three paragraphs and the carcinoid flow diagram of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C09-BD2

Brain dump 2 · integration and application: A flushing patient with a murmur that changes with breathing

A competent response contains

· Flushing and bronchospasm with wheezing come from kinins, histamine, and tachykinins; watery diarrhea and cramps come from serotonin.

· The holosystolic murmur at the left lower sternal border that increases on inspiration is tricuspid regurgitation; the systolic ejection murmur at the left upper sternal border is pulmonic stenosis.

· It increases on inspiration because inspiration raises venous return to the right heart and therefore the regurgitant flow.

· Large jugular v waves, a pulsatile liver, edema, and ascites follow from tricuspid regurgitation and right heart failure.

· The rash and memory change are pellagra: tryptophan is diverted to serotonin synthesis, leaving too little for niacin, and the deficiency gives dermatitis, diarrhea, and dementia.

· Urinary 5-hydroxyindoleacetic acid over 24 hours is high; it is the monoamine oxidase metabolite of serotonin.

· The tumour must have metastasised to the liver, because liver metastases release products into the hepatic veins and bypass hepatic monoamine oxidase.

· His left-sided valves should be spared, because pulmonary endothelium takes up and degrades serotonin before the blood reaches the left heart.

· The patient with a bronchial carcinoid and no liver metastases develops left-sided lesions, because a bronchial tumour drains into the pulmonary veins and delivers its products to the left atrium without passing the liver or the pulmonary capillary bed, while the right heart receives systemic venous blood with little serotonin.

Distinctions that matter

· Different mediators produce different symptoms: serotonin drives the diarrhea and the valve plaques, while kinins, histamine, and tachykinins drive the flushing and wheeze.

· A right-sided murmur that increases on inspiration is distinguished from a left-sided murmur by that respiratory change alone.

Common misconceptions

· Explaining the flushing by serotonin alone. Kinins, histamine, and tachykinins are responsible.

· Expecting the valve lesion to be on the side where the tumour sits. The route of venous drainage decides it.

Where it is taught: Section C09, symptom and sign paragraph and the stop and predict box of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C09-Q03

Board-style question: Valve disease in a patient with liver metastases

Correct answer: D. Tricuspid regurgitation with pulmonic stenosis from fibrous plaques that fix the leaflets

Why it is right: Serotonin that bypasses hepatic monoamine oxidase through liver metastases reaches the tricuspid and pulmonic valves first and drives plaque formation through 5-HT2B receptors, fixing the tricuspid leaflets open to give regurgitation while the retracted pulmonic cusps give stenosis, often with regurgitation.

A is wrong: Warty vegetations on both surfaces of the leaflets with mitral regurgitation describe Libman-Sacks endocarditis in lupus, not a serotonin-driven plaque.

B is wrong: Heaped calcified cusps with unfused commissures describe age-related calcific aortic stenosis, and the left-sided valves here are normal.

C is wrong: Commissural fusion with a fish-mouth orifice describes chronic rheumatic mitral stenosis, which follows streptococcal infection rather than a neuroendocrine tumour.

A competent response contains

· Flushing, wheeze, and diarrhea with a raised urinary 5-hydroxyindoleacetic acid and an ileal tumour with liver metastases establish carcinoid syndrome with cardiac involvement.

· Serotonin reaching the right heart acts on 5-HT2B receptors of valve interstitial cells, producing pearly fibrous plaques that fix the tricuspid leaflets open and thicken and retract the pulmonic cusps.

Distinctions that matter

· Carcinoid plaques lie on the surface and fix the leaflets, which is why tricuspid regurgitation predominates with pulmonic stenosis alongside.

Common misconceptions

· Expecting left-sided lesions. Pulmonary endothelium takes up and degrades serotonin before the blood reaches the left heart.

Where it is taught: Section C09 of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C10-BD1

Brain dump 1 · foundational retrieval: Two kinds of prosthesis and the ways each one fails

A competent response contains

· Mechanical valves are usually bileaflet pyrolytic carbon, last decades, and click crisply, but their foreign surface and abnormal flow make them highly thrombogenic.

· Bioprosthetic valves are porcine valves or bovine pericardium fixed in glutaraldehyde, which crosslinks collagen, kills the cells, and lowers antigenicity; the tissue is far less thrombogenic but nonviable and unable to repair itself, so structural deterioration commonly occurs within about 10 to 20 years and faster in younger patients.

· Transcatheter valves are bioprosthetic tissue mounted on a metal frame.

· Virchow's triad on a mechanical valve: the foreign surface adsorbs plasma proteins and activates platelets and the contact pathway (endothelial or surface injury), flow around the hinges creates zones of stasis and high shear (abnormal flow), and hypercoagulability or interrupted anticoagulation completes it.

· Mitral prostheses carry greater thrombotic risk than aortic ones because mitral flow is slower and the left atrium is often large and fibrillating.

· Thrombus can immobilise a leaflet, giving a new murmur, muffled clicks, heart failure, and shock, or it can embolise.

· Pannus is a slower fibrous ingrowth from the sewing ring that narrows the orifice over months to years.

· High-velocity turbulent flow fragments red cells, worst through a paravalvular leak, which is a gap between the sewing ring and the tissue.

· Laboratory findings: schistocytes including helmet cells from shear fragmentation, high lactate dehydrogenase and unconjugated bilirubin from released cell contents and heme breakdown, low or absent haptoglobin because it is consumed binding free hemoglobin, reticulocytosis from marrow compensation, and hemoglobinuria from filtered free hemoglobin.

· Tubular cells store filtered hemoglobin iron as hemosiderin and shed it, so chronic urinary iron loss can cause iron deficiency.

· Bioprosthetic tissue undergoes dystrophic calcification as calcium binds the phospholipids of devitalised cells and altered collagen, faster in children and young adults, with collagen fatigue from repeated flexing adding to it, so cusps stiffen giving stenosis or tear giving regurgitation, sometimes suddenly.

Distinctions that matter

· Mechanical valves fail by thrombosis and shear; bioprosthetic valves fail by calcification and tearing of nonviable tissue.

· Thrombosis presents over hours to days while pannus narrows the orifice over months to years.

Common misconceptions

· Assuming a bioprosthetic valve needs no monitoring because it is less thrombogenic. Its tissue cannot repair itself and deteriorates structurally.

· Expecting the hemolysis to be immune. It is mechanical destruction by shear.

Where it is taught: Section C10, first two paragraphs, the hemolysis paragraph, and the calcification paragraph of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C10-BD2

Brain dump 2 · integration and application: Read the panel and name the complication

A competent response contains

· This is mechanical intravascular hemolysis, most likely from a paravalvular leak, a gap between the sewing ring and the tissue through which a high-velocity jet shears red cells apart.

· Schistocytes and helmet cells mark the shearing itself; high lactate dehydrogenase marks release of cell contents; raised unconjugated bilirubin marks heme breakdown; absent haptoglobin marks its consumption binding free hemoglobin; reticulocytosis marks marrow compensation; hemoglobinuria and hemosiderinuria mark filtration of free hemoglobin and storage of its iron in tubular cells that are then shed.

· A negative direct antiglobulin test shows the destruction is mechanical rather than antibody-mediated, and a normal platelet count separates this from disseminated intravascular coagulation, thrombotic thrombocytopenic purpura, and hemolytic uremic syndrome, which also produce schistocytes; autoimmune hemolysis is the immune process the antiglobulin test excludes.

· She can become iron deficient because iron is lost continuously in the urine as hemoglobin and shed hemosiderin, so it is not recycled as it would be in extravascular hemolysis.

· Prosthetic valve endocarditis begins at the sewing ring and often spreads into the annulus, causing ring abscess, dehiscence, paravalvular leak, and heart block.

· Early disease, within about the first year and most often the first 2 months, reflects perioperative contamination, and the expected organism is a coagulase-negative staphylococcus, especially Staphylococcus epidermidis, alongside S. aureus, gram-negative bacilli, and Candida.

· It depends on a polysaccharide biofilm that binds synthetic material and shields the organism from neutrophils, complement, and antibiotics.

· Laboratory features of that organism: gram-positive cocci in clusters, catalase positive, coagulase negative, and novobiocin sensitive.

· Late disease after valve replacement resembles native valve endocarditis in its organisms.

Distinctions that matter

· Mechanical and microangiopathic hemolysis both give schistocytes, and the platelet count plus the valve finding separate them.

· Early and late prosthetic valve endocarditis differ by organism, because early disease reflects contamination at surgery.

Common misconceptions

· Reading schistocytes as proof of a microangiopathy. A prosthetic jet shears cells just as effectively, with a normal platelet count.

· Assuming hemolysis always raises iron stores. Intravascular hemolysis loses iron in the urine and can cause deficiency.

Where it is taught: Section C10, hemolysis paragraph, the hemolysis flow diagram, and the prosthetic valve endocarditis paragraph of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C10-Q04

Board-style question: Anemia three years after valve replacement

Correct answer: B. Shear fragmentation of red cells by a high-velocity jet through a paravalvular leak

Why it is right: The echocardiogram shows a gap between the sewing ring and the annulus with a high-velocity jet, and shear stress through that leak fragments red cells inside the circulation, which accounts for the helmet cells, the high lactate dehydrogenase and bilirubin, the absent haptoglobin, and the hemoglobin and hemosiderin in the urine.

A is wrong: Warm autoantibody hemolysis would give a positive direct antiglobulin test, and hers is negative; antibody-coated cells are also cleared in the spleen rather than fragmented.

C is wrong: Platelet consumption with fibrin strands in the microvasculature would lower the platelet count, and hers is 240,000/mm3.

D is wrong: Hepcidin-driven iron restriction causes anemia without hemolysis, so it would not produce fragmented cells, absent haptoglobin, or hemoglobinuria.

A competent response contains

· A high-velocity jet through a paravalvular leak shears red cells, giving schistocytes and helmet cells with high lactate dehydrogenase, raised unconjugated bilirubin, absent haptoglobin, and hemoglobinuria with hemosiderinuria.

· A negative direct antiglobulin test shows the destruction is mechanical, and a normal platelet count excludes the consumptive microangiopathies.

Distinctions that matter

· Intravascular hemolysis loses iron in the urine, which is why chronic iron deficiency can follow despite ongoing red cell destruction.

Common misconceptions

· Taking schistocytes as evidence of a microangiopathy without checking the platelet count and the prosthesis.

Where it is taught: Section C10 of the CV-18-D Valvular Heart Diseases notes.